

Assessment 1 - Data Analysis and Classification for E-Commerce Sales Dataset

About the Project and Dataset

This dataset provides detailed insights into Amazon sales data, including SKU Code, Design Number, Stock, Category, Size and Color, to help optimize product profitability

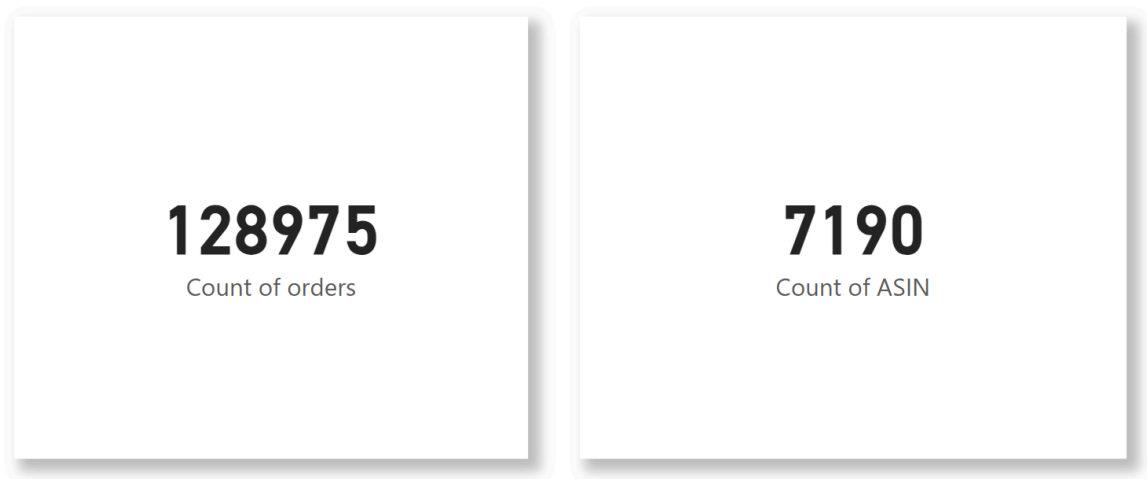
Workflow of this project

This project is processed basically following the sequenced operations

1. Choose the target dataset from
<https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data>
2. Modify original dataset to be transformed as suitable for later data mining
 - a. Get rid of missing values
 - b. Remove outliers
 - c. Remove insignificant attributes
3. Use Weka for data mining and make assumption for the pattern of dataset
 - a. Visualize the dataset to find correlations among attributes
4. If necessary, use Power BI to prove and reevaluate the hypothesis from Weka
5. Utilize four classifiers and get training result of dataset

Description of selected dataset

Using Power BI and get the count of records of original dataset



This dataset include details of orders on Amazon basically from India. By this dataset analyzed, we can estimate possible correlations among these attributes (for example, is there correlation between shipped city and date of order) therefore we could find our Amazon's potential strategies and data evidence to enhance profitability on our business

Reduction of attributes

Their are 24 types of attributes in the original data set. Here are sample of one record.

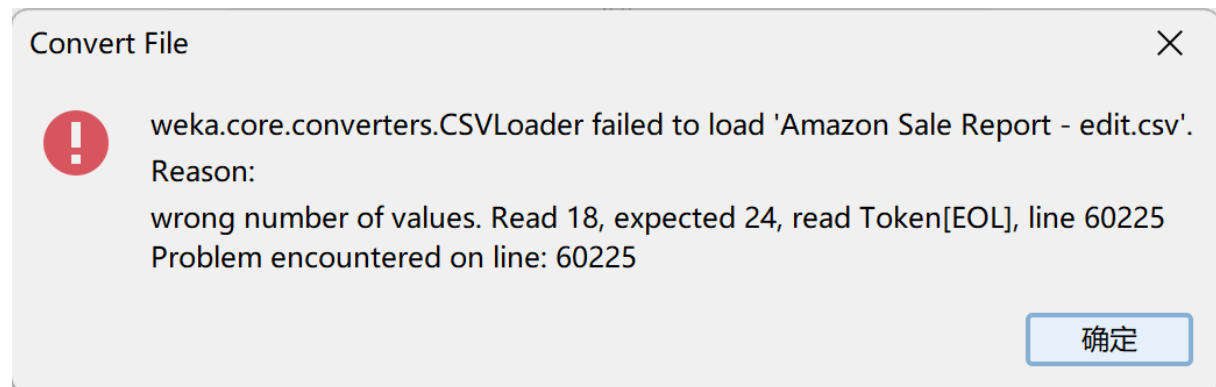
Explanations for some of the attributes

- **Category** : Type of product. (String)
- **Size** : Size of the product. (String)
- **Date** : Date of the sale. (Date)
- **Status** : Status of the sale. (String)
- **Fulfilment** : Method of fulfilment. (String)
- **Style** : Style of the product. (String)
- **SKU** : Stock Keeping Unit. (String)
- **ASIN** : Amazon Standard Identification Number. (String)
- **Courier** Status: Status of the courier. (String)
- **Qty** : Quantity of the product. (Integer)
- **Amount** : Amount of the sale. (Float)
- **B2B** : Business to business sale. (Boolean)
- **Currency** : The currency used for the sale. (String)

Data Preparation

Convert CSV file into ARFF file

An error occurs that seems because of invalid syntax 



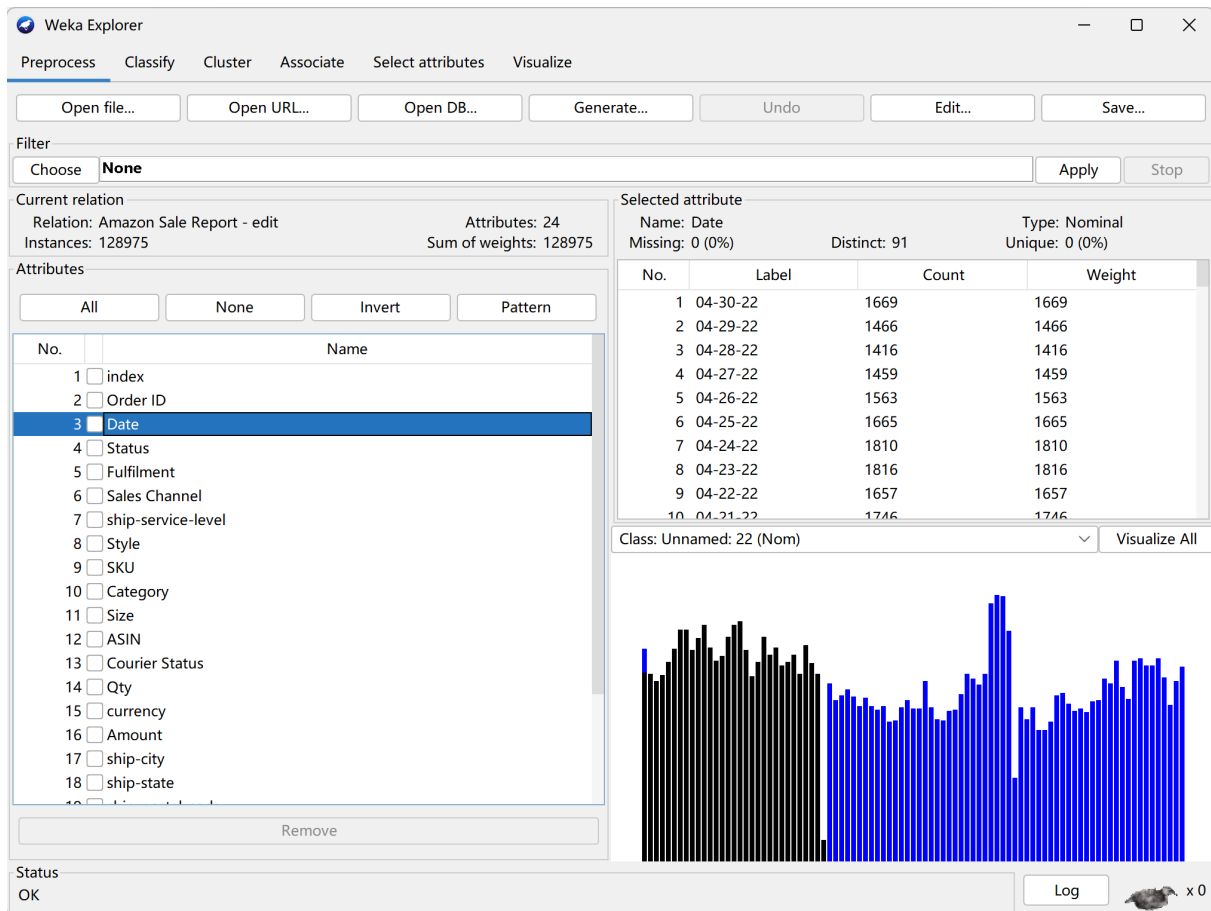
Therefore, for each string containing syntax , I added  on its both sides

.Near Laurel's international school,

>>>>>

"Near Laurel's international school"

Then, the CSV successfully converted into ARFF and be loaded onto Weka



Remove missing values and outliers

Viewer

Relation: Amazon Sale Report - edit

1: Index	8: Style	9: SKU	10: Category	11: Size	12: ASIN	13: Courier Status	14: Qty	15: currency	16: Amount	17: ship-city	18: ship-state	19: ship-postal-code	20: ship-postal-code
Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Numeric	Nominal	Numeric	Nominal	Nominal	Numeric	Nominal
JNE38...	JNE38...	Western Dr...	XXL	B09TH5...			0.0	INR	734.29	VIJAYAWA...	ANDHRA P...	520012.0	IN
J0381	J0381...	Set	3XL	B09M6...			0.0	INR	1023.21	VIJAYAWA...	ANDHRA P...	520012.0	IN
JNE37...	JNE37...	Western Dr...	3XL	B09SDY...			0.0	INR	689.52	VIJAYAWA...	ANDHRA P...	520012.0	IN
PJNE3...	PJNE3...	kurta	6XL	B09LD4...	Shipped		1.0	INR	486.0	BENGALURU	KARNATAKA	560036.0	IN
PJNE2...	PJNE2...	kurta	6XL	B09LD4...	Shipped		1.0	INR	431.0	BENGALURU	KARNATAKA	560036.0	IN
JNE33...	JNE33...	kurta	L	B081X4...	Shipped		1.0	INR	471.0	MHOW	MADHYA P...	453441.0	IN
SET227	SET22...	Set	L	B08P1D...	Shipped		1.0	INR	571.0	Allahabad	UTTAR PRA...	211016.0	IN
SET331	SET33...	Set	XXL	B09NQ...	Shipped		1.0	INR	597.0	ERNAKUL...	KERALA	682020.0	IN
JNE32...	JNE32...	kurta	M	B07R5X...	Unshipped		1.0	INR	348.0	HYDERABAD	TELANGANA	500015.0	IN
JNE36...	JNE36...	kurta	M	B091Q8...	Shipped		1.0	INR	318.0	HOSKOTE	KARNATAKA	562114.0	IN
JNE36...	JNE36...	Top	L	B0943J...			0.0			HOSKOTE	KARNATAKA	562114.0	IN
PJNE3...	PJNE3...	kurta	4XL	B09M7...	Shipped		1.0	INR	692.0	BAGALKOT	KARNATAKA	587101.0	IN
JNE37...	JNE37...	kurta	3XL	B09K3Z...	Shipped		1.0	INR	362.0	HYDERABAD	TELANGANA	500016.0	IN
JNE37...	JNE37...	kurta	3XL	B09K3T...	Shipped		1.0	INR	345.0	HYDERABAD	TELANGANA	500016.0	IN
JNE38...	JNE38...	Western Dr...	XXL	B09RK6...	Shipped		1.0	INR	677.0	RAJPUR S...	WEST BENG...	700148.0	IN
J0135	J0135...	Set	XL	B08QGJ...	Shipped		1.0	INR	591.0	JAMMIKU...	TELANGANA	505122.0	IN
SET293	SET29...	Set	3XL	B09K3C...	Shipped		1.0	INR	736.0	MUMBAI	MAHARASH...	400018.0	IN
SET377	SET37...	Set	XXL	B09TZT...	Shipped		1.0	INR	1238.0	MATHURA	UTTAR PRA...	281003.0	IN
JNE37...	JNE37...	kurta	M	B09K3X...	Shipped		1.0	INR	499.0	BENGALURU	KARNATAKA	560068.0	IN
J0094	J0094...	kurta	M	B089G1...	Shipped		1.0	INR	568.0	BENGALURU	KARNATAKA	560068.0	IN
J0012	J0012...	Set	XL	B0894...	Shipped		1.0	INR	1140.0	THANE	MAHARASH...	400612.0	IN
SET331	SET33...	Set	M	B09NQ...	Unshipped		1.0	INR	597.0	THANE	MAHARASH...	400608.0	IN
J0119	J0119...	Top	M	B08RY...	Shipped		1.0	INR	0.0	Bangalore	KARNATAKA	560016.0	IN

Add instance Undo OK Cancel

1. Blank value on currency and amount do not make sense on real case of sales on Amazon, therefore, they are to be removed from the dataset

Filter	
Choose	RemoveWithValues -S 0.0 -C 15 -L 1 -M

2. Index and Order ID are unique or auto-increasing, it's unnecessary to include these attributes since they are not considered as features related to the case itself, furthermore it could be overfitting and does not contribute to data prediction

No.	Name
1	☒ index
2	☒ Order ID
3	☐ Date
4	☐ Status
5	☐ Fulfilment
6	☐ Sales Channel
7	☐ ship-service-level
8	☐ Style
9	☐ SKU
10	☐ Category
11	☐ Size
12	☐ ASIN
13	☐ Courier Status
14	☐ Qty
15	☐ currency
16	☐ Amount
17	☐ ship-city
18	☐ ship-state
19	☐ ship-postal-code
20	☐ ship-country
21	☐ promotion-ids
22	☐ B2B
23	☐ fulfilled-by
24	☐ Unnamed: 22

Remove

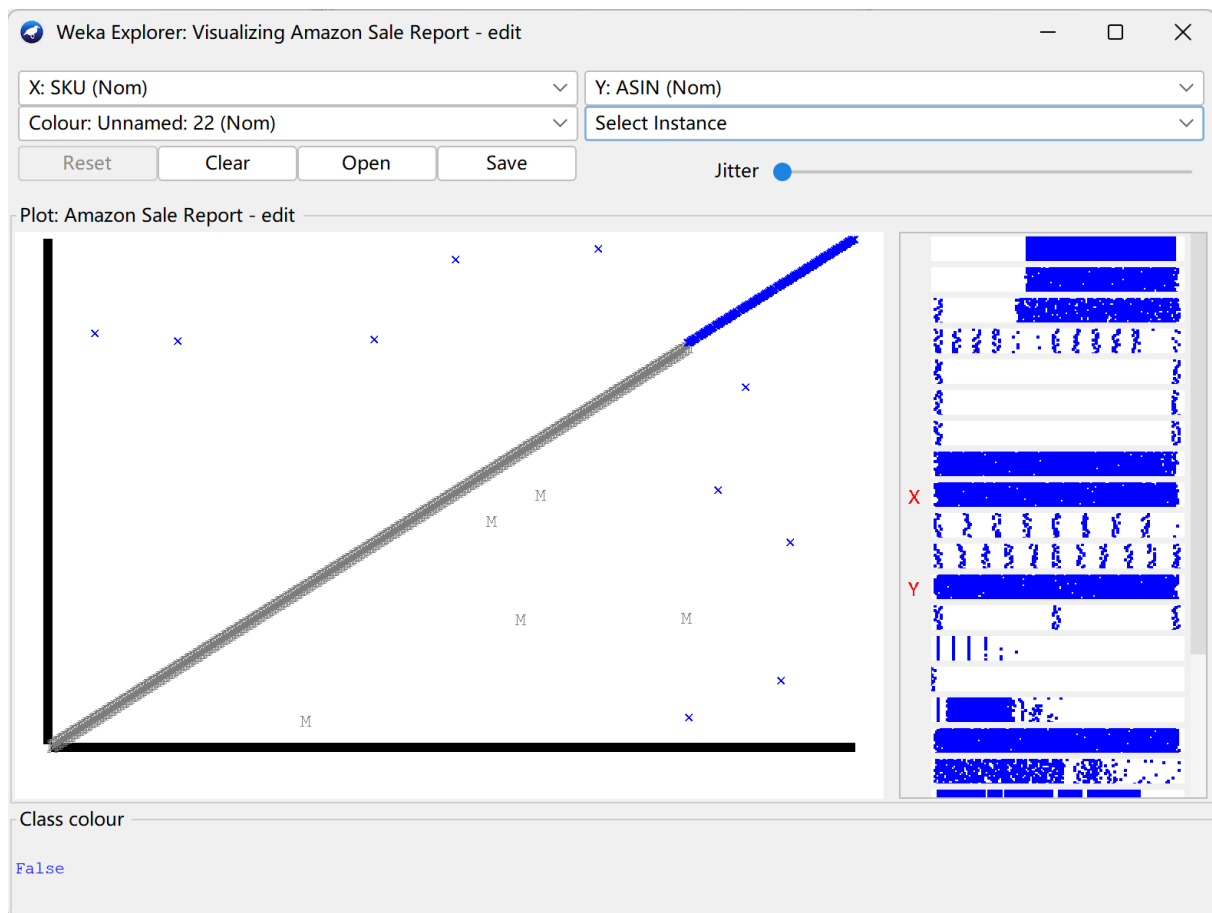
Data visualization and analysis

There are eight correlations with scatter plot in Weka.

- ASIN & SKU
- Ship city & Date
- Promotion ID & Date
- Category & Amount
- Status & Courier status
- Status & Quantity
- Status & Fulfilment
- Fulfilment & Ship service level

1. Correlation between ASIN and SKU

Go to Visualize panel and set axis to observe ASIN and SKU which is shown as blow



Analysis to the chart

This chart illustrates values in plot forms a straight diagonal line, which indicate that at most every SKU has its own ASIN respectively, which means their Correlation as a potential **one-to-one direct** correspondence.

However there are also points scatter around this line, which means there are couple set of SKU and ASIN that do not strictly follow one-to-one correspondence. That is to say, one SKU may map to two or more ASINs or vice versa, while this situation is minority.

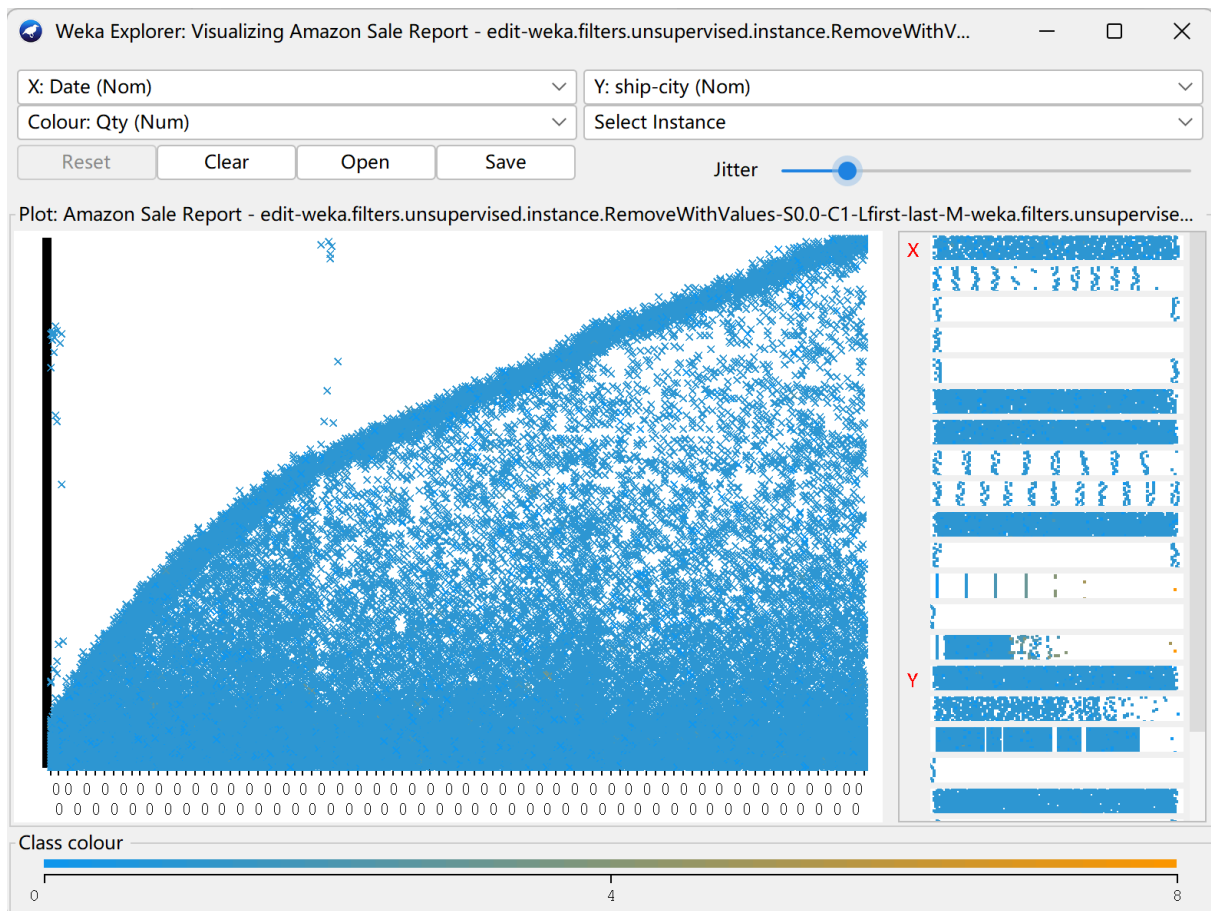
That phenomenon may because Amazon distribute different ASINs for a single SKU in different region (assumption).

Confirmation from Power BI

SKU	count of ASIN	ASINs
J0027-SET-XS	2	B08B3YZN3J, B08B4T6KJK
J0091-TP-XL	2	B092D2LWSQ, B092ZXCCNJ
J0157-DR-S	2	B0982ZDZ8K, B098P3LXT1
J0212-DR-L	2	B0982Z8HPJ, B098P4Z9WZ
JNE3713-TP-N-XXL	2	B091GV1PQR, B08ZHGDW8
AN201-RED-M	1	B091GP22TP
AN201-RED-XL	1	B091GLZF98
AN201-RED-XXL	1	B091GP61BH
AN202-ORANGE-M	1	B091GMKCQP
AN202-ORANGE-S	1	B091GP4R8Y
AN202-ORANGE-XXL	1	B091GK5Y64
AN203-MAROON-XXL	1	B091GPQB62

This prove the Correlation as I assumed by the plot using Weka that SKU and ASIN are mostly following one-to-one direct Correlation with a few exceptions

2. Correlation between Ship city and Date



Analysis to the chart

This chart illustrates a rough diagonal line when upscaling jitter, which indicates a **one-to-one Correlation** between ship-city and date. That is to say, there are periods for cities that frequent sales activities but not consistently keep the frequency over all. For instance, customers living in A make unusually frequent orders during March and others living in B do so during April. The regional pattern may be led by **regional promotion**.

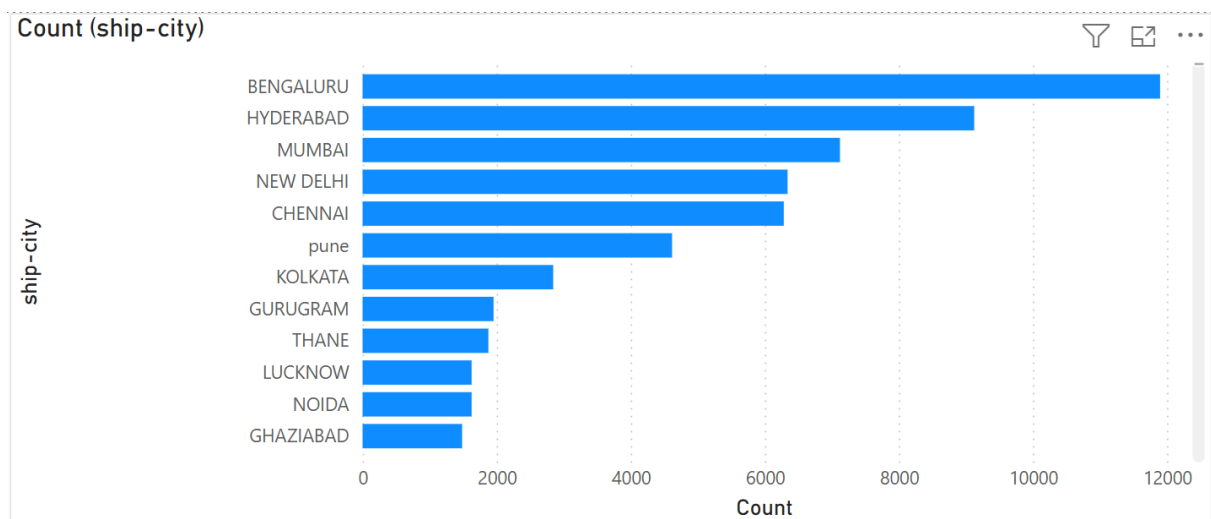
Additionally, the density for ship-city over time is not flat - some area is more dense than others. There are two features about density - Extremely low density at top-left area and high density at bottom-right. Which means a huge regional gap in sales frequency - some cities only have few purchases while some have high loyal customers.

There are several potential reasons:

- The high **regional gap** in economy and population in India (for most data taken from India)
- Regional gap in acceptance to Amazon (residents in different cities tend to use different shopping platforms)

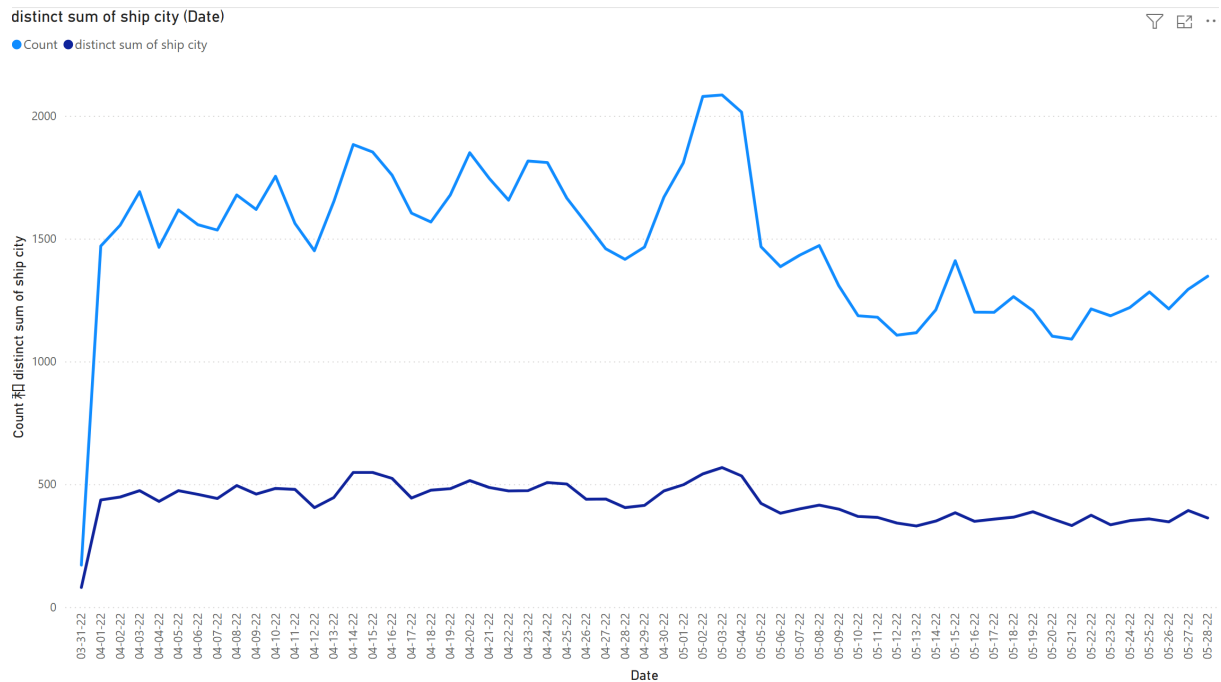
Commonly, the former explanation is more realistic.

Confirmation from Power BI



This chart shows the regional gap in orders count that Bengaluru gets 11,898 orders while the city with the no.12 orders - Ghaziabad gets 1,485.

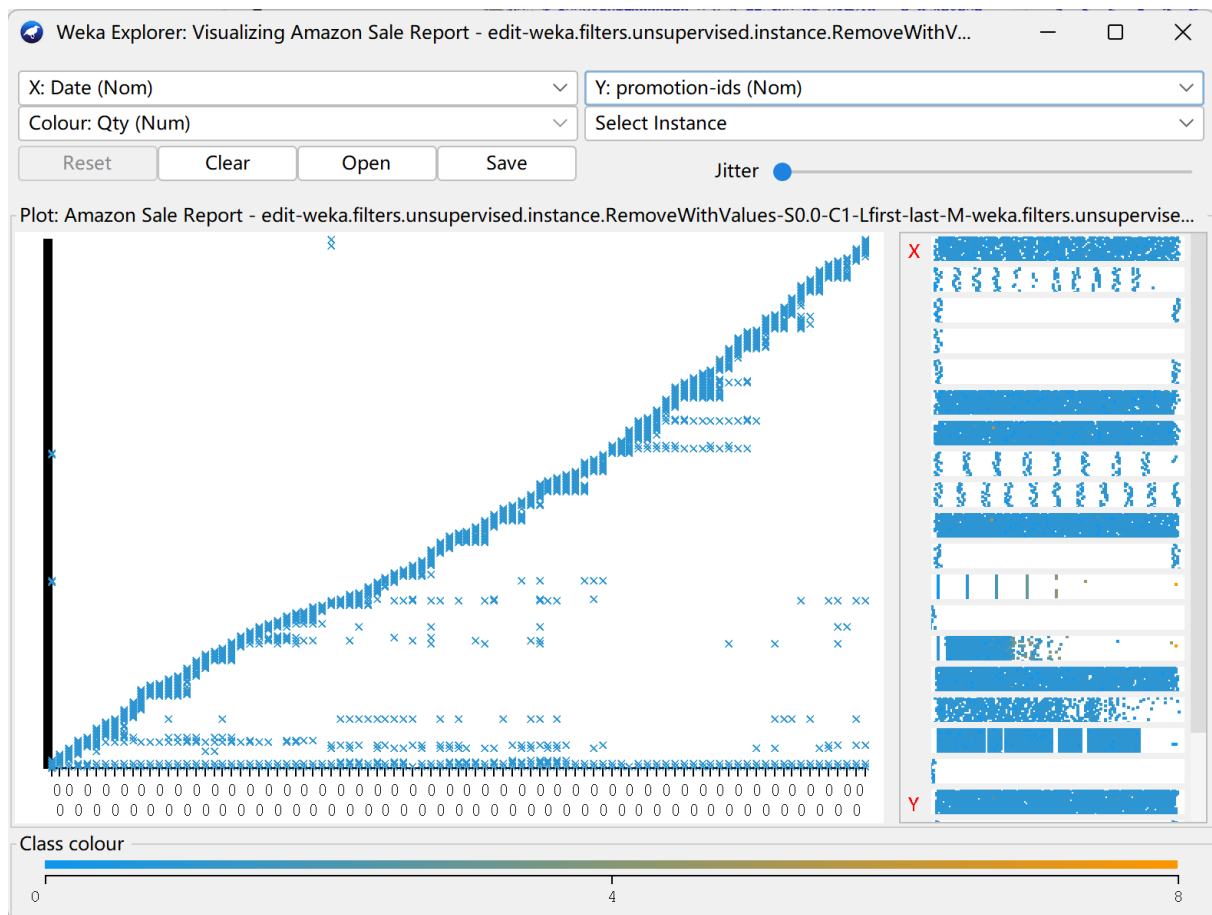
According to Wikipedia, Bengaluru and Ghaziabad have respectively 8,448K and 17,29K (data from 2011), that means each resident in Bengaluru makes 60% more Amazon purchases than one in Ghaziabad.



The lines for distinct sum of city and sum of city against time show no prominent increase or fall but only fluctuations except from 2022-03-31 when data is extremely low.

It is easy to misunderstand when looking at the scatter plot in Weka and decide there is a increasing trend of variety of ship-city as there is a uptrend on that. However scatter plot doesn't define its nominal variables (in this case date) following any order neither in ASE nor DESC but depending on scattering pattern.

3. Correlation between Promotion ID and Date



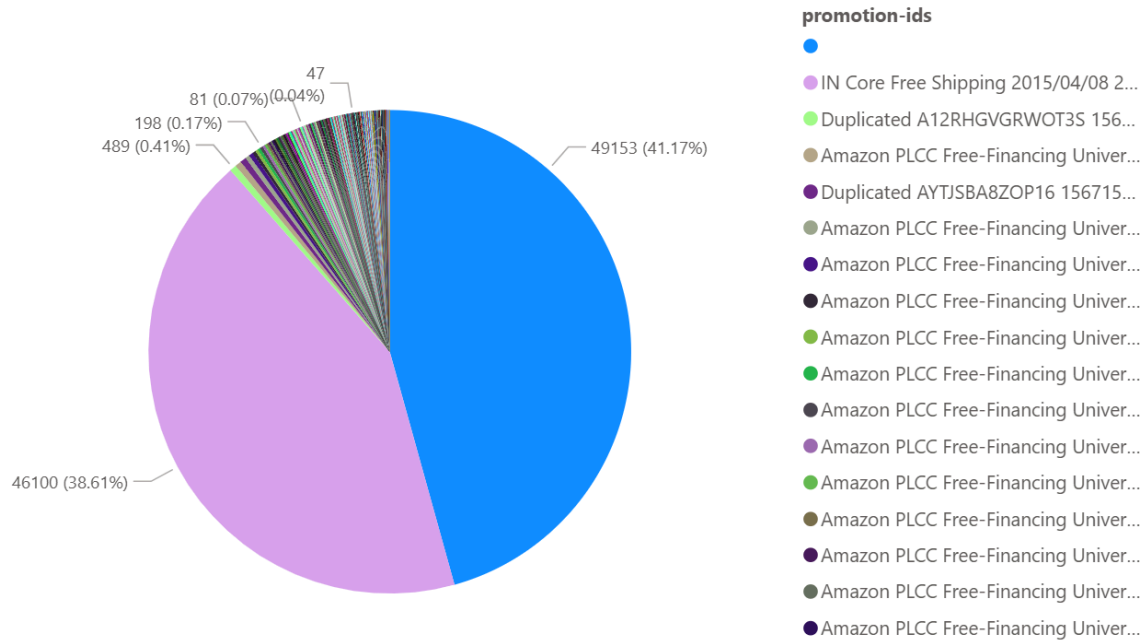
Analysis to the chart

This chart shows a rough diagonal line and a parallax line across the duration of all dates, which means the promotion id has two-fold kinds of feature - one as a roughly **one-to-one direct Correlation** against date and another one as commonly used everyday.

In conclusion, most promotion ids only exist for a few days, while a few ids remains for a long period, furthermore there are very rare ids could be used almost everyday.

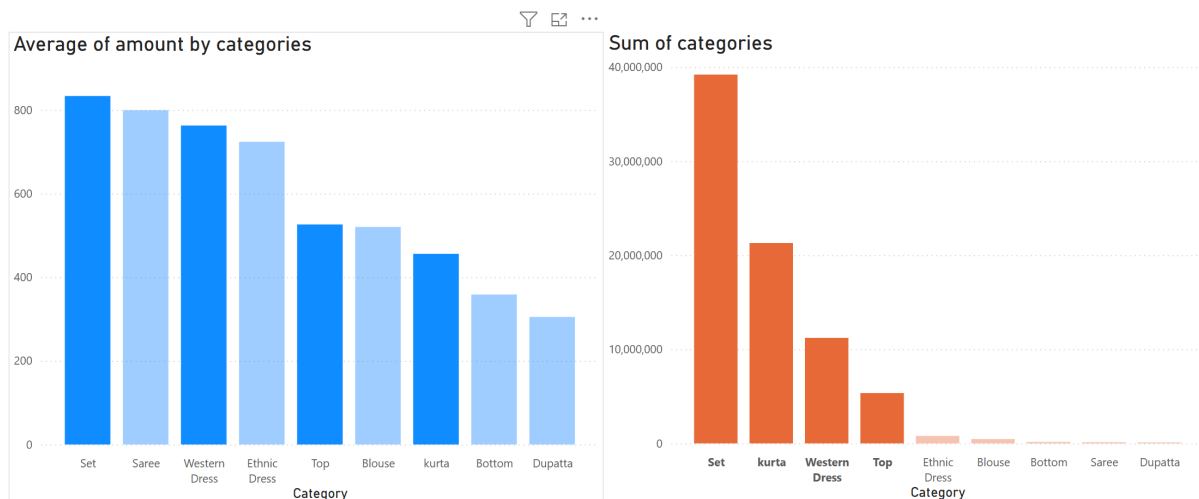
Confirmation from Power BI

Count (promotion-ids)



Additionally, with the pie chart created by Power BI which shows the empty value share the biggest part with 41% and subsequently followed by **IN Core Free Shipping 2015/04/08 23-48-5-108** with 39%, which must be the main part of the lying parallax line of values.

4. Correlation between Category and amount over period

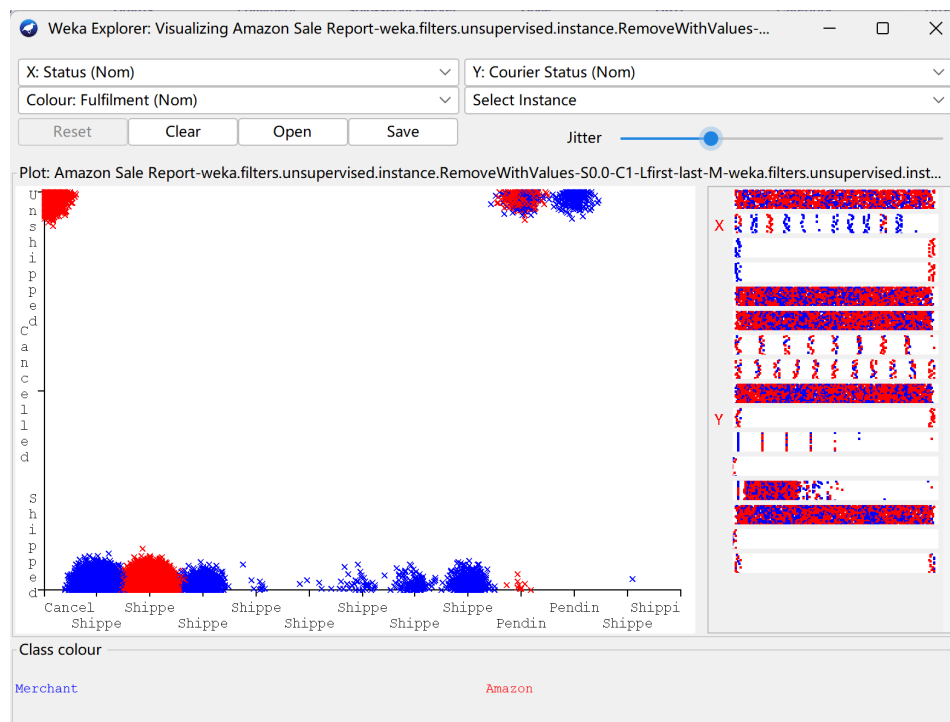


These two charts illustrate count and average of amount for every categories. By selecting top 4 often categories at the right chart which is **Set**, **Western Dress**, **Top** and lastly **Kurta**.

It is evident as the highlighted categories at the left chart that the average amount for these categories are ranked as mostly the same as they were analyzed from Weka.

By the way, it is interesting to find that both the average amount and count for good of **Set** are highest. That is why it is obviously illustrated with the largest area of 'blue' as it represents that goods of **Set** are the best-selling category over period.

5. Correlation between Status and Courier Status



Analysis to the chart

The chart illustrate correlation between Status and Courier status, which respectively related to annotation to status of orders and status of fulfilment.

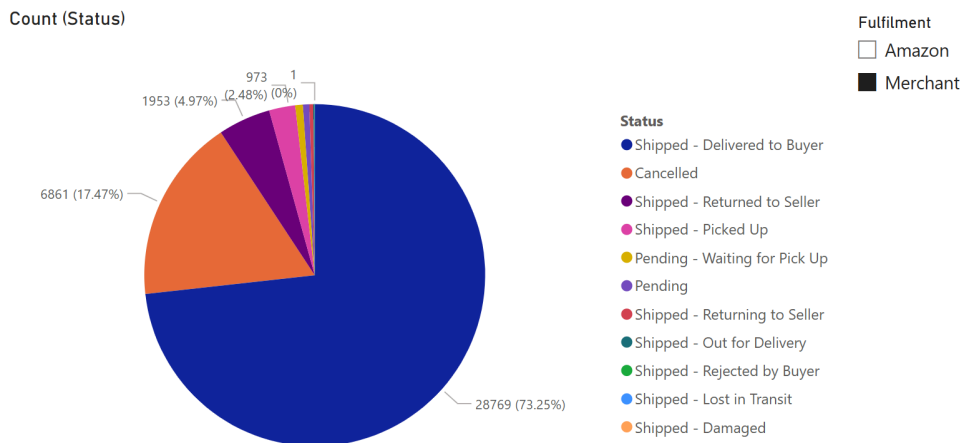
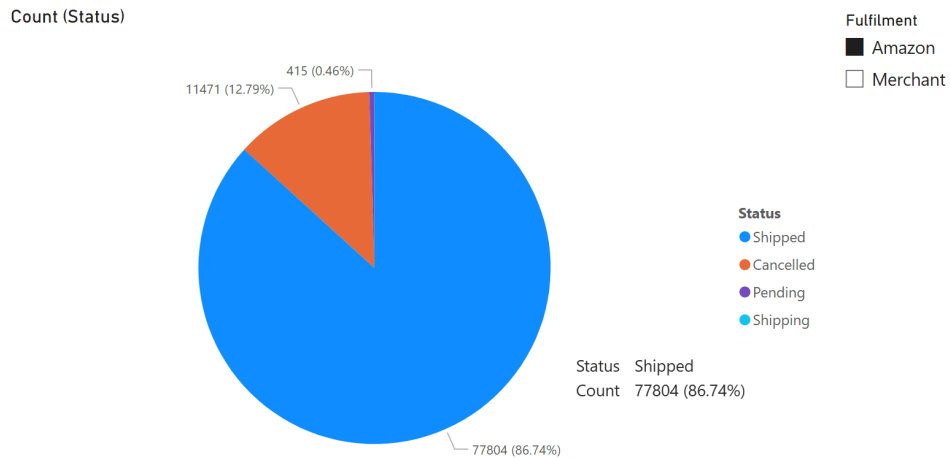
They are both nominal with a few elements so I upscale jitter to get better vision on scatter plot, where they follow a clear relations:

- If an order has courier status as **Cancelled** , the status is to be **Unshipped** and it is to be fulfilled by Amazon
- Those with Amazon fulfilment only have status of **Shipped** , **Cancelled** and **Pending** , while those with Merchant fulfilment have more detailed status shipped status instead of **Shipped** and other to **Pending** , there is **Pending - Waiting for Pick Up** in status.
- There are no **Cancelled** in Courier status, this is because those with it get blank value in currency and amount which have been removed during data preparation stage.
- If an order has **Pending** as its status, it often means **Unshipped** as its courier status. However, this is not an exact correlation - sometime **Pending** status together with **Shipped** as courier status.

Confirmation from Excel

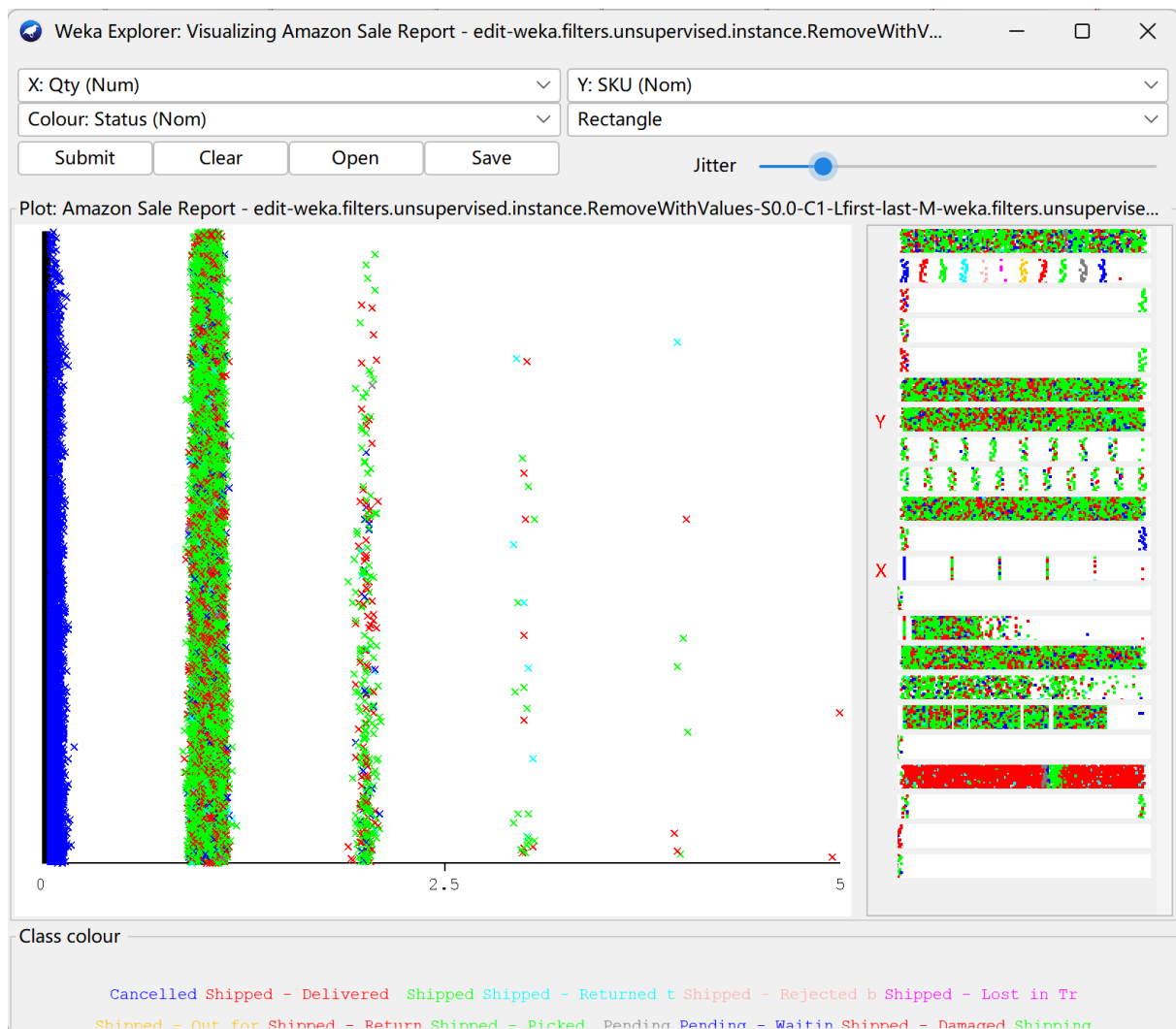
B	K	L	M	N
Status	Courier Status	Qty	currency	Amount
Cancelled			0 INR	647.62
Cancelled			0 INR	753.33
Cancelled	Cancelled		0	
Cancelled			0 INR	570.48
Cancelled			0	
Cancelled	Cancelled		0	
Cancelled			0 INR	1105.36
Cancelled	Cancelled		0	
Cancelled	Cancelled		0	
Cancelled	Cancelled		0	
Cancelled	Unshipped		1 INR	791
Cancelled	Unshipped		1 INR	807
Cancelled	Unshipped		1 INR	807
Cancelled	Cancelled		0	
Cancelled	Cancelled		0	
Cancelled	Cancelled		0	
Cancelled	Cancelled		0	

Records with CANCELLED courier status confirmed to get empty **currency** and **amount**



These two pie charts prove Amazon and Merchant fulfilments get exclusive approaches to mark status - Amazon get simpler divisions of status while Merchant ones go with more detailed categories.

6. Correlation between Status and Quantity

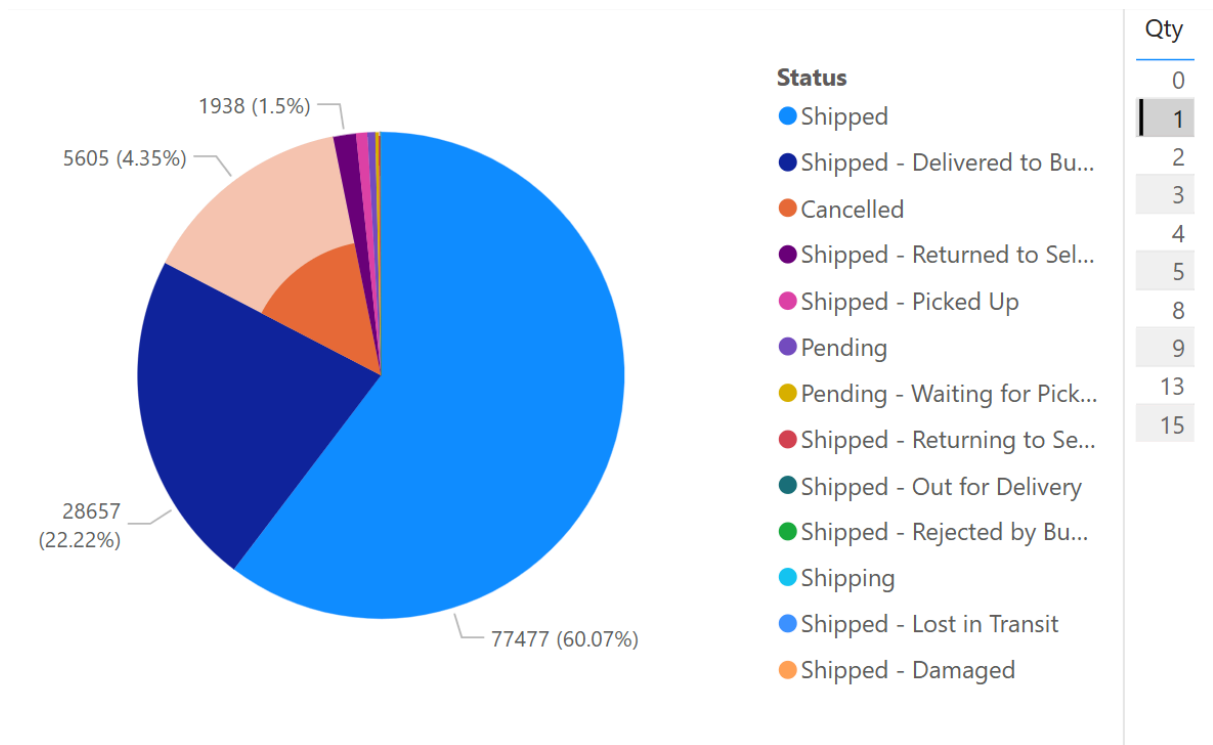


Analysis to the chart

It is evidence that the variety of SKU shrinks dramatically as the quantity grows - most purchase involves single product, while there a considerable number of cancelled records all of which get 0 in quantity.

It is clear that orders with **Shipped** and **Shipped - Delivered** get the largest share on non-cancelled orders due to the dominating areas of green and red, which means most orders was successfully delivered with several goods returned. Notably, orders with one Qty share the most proportion among all and cover almost every SKU.

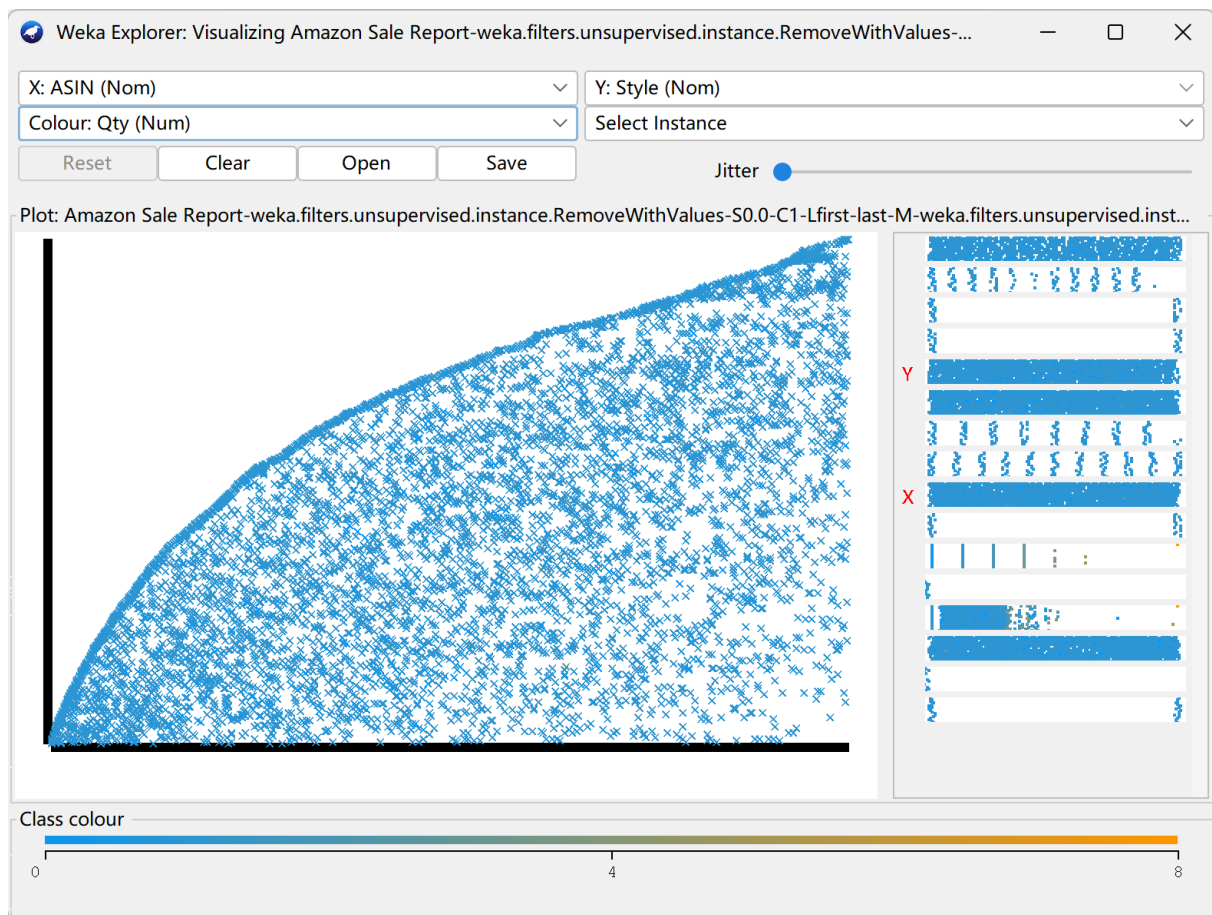
Confirmation from Power BI



The pie chart shows the share of all statuses filtered by condition with one Qty. Single Qty orders are the majority for every sort of status, except from **Cancelled** which mainly occur in those with 0 Qty.

Shipped and **Shipped - Delivered to Buyer** hold the largest share at respectively 60.07% and 22.22%, which justifying our assumption from Weka

7. Correlation between ASIN and Style



Analysis to the chart

This chart illustrates two features - high density at bottom-right and a rough diagonal dense line.

Dense at bottom-right

Points mostly scattered at bottom-right means ASIN is more various than Style, which means one Style could be mapped to several ASINs.

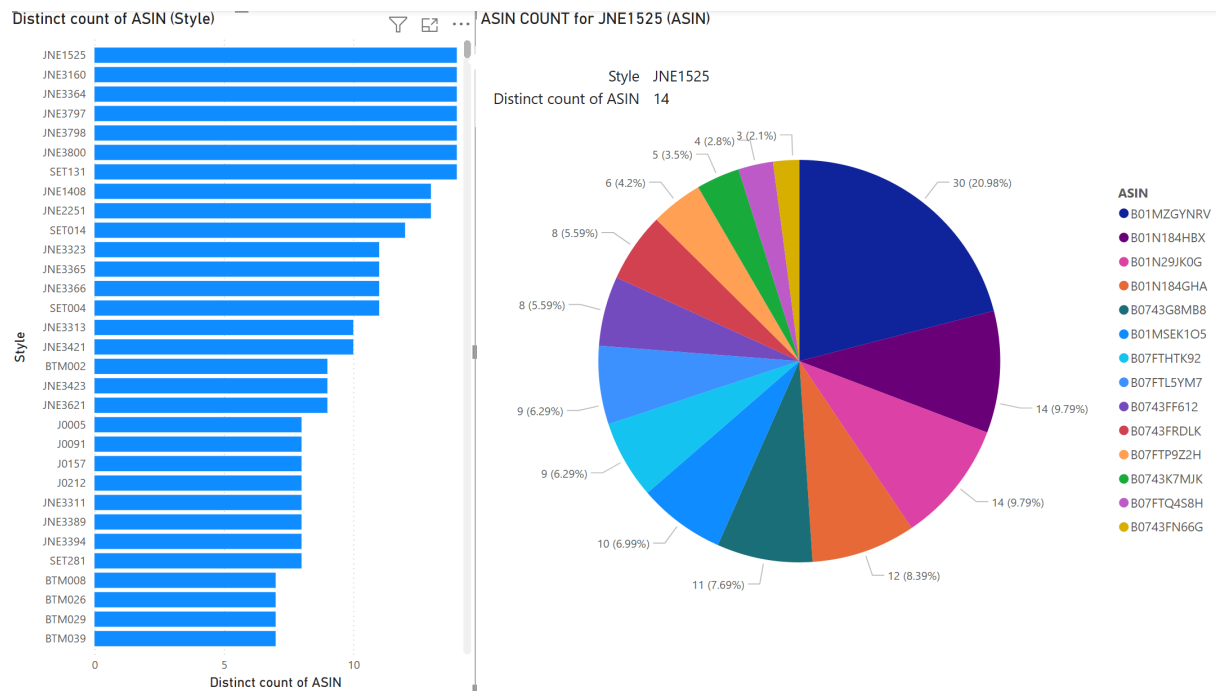
Dense diagonal line

There are many combinations of ASIN and Style with positive relation, which means there are always ASINs more popular than others that are sharing the same style.

For instance, style A gets ASINs of 'a', 'b', 'c' to combine with, and 'a' is much more popular than 'b' and 'c'.

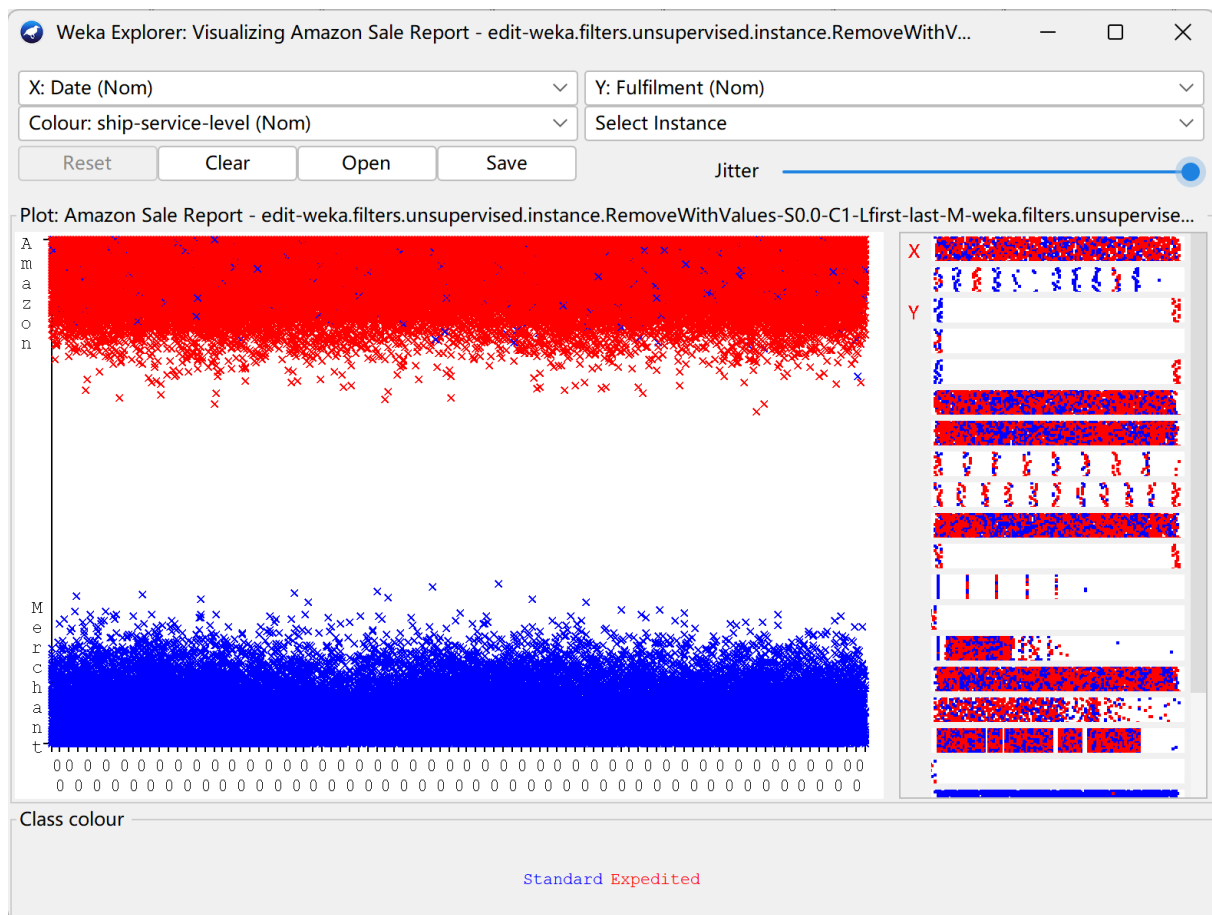
This reveals a business phenomenon that customers tend to have similar preferences. Shopkeepers should pay attention to import more models of more popular ones to reduce stock redundancy.

Confirmation from Power BI



It is evident shows on left chart that there are styles with many possible ASINs (for instance JNE1525 has 14 kinds of ASINs). While the pie chart on the right shows proportion of count of ASINs of JNE1525, where ASIN of **B01MZGYNRV** is clearly the most popular ASIN (30 purchases in total 143)

8. Correlation between Fulfilment and Ship-Service-Level



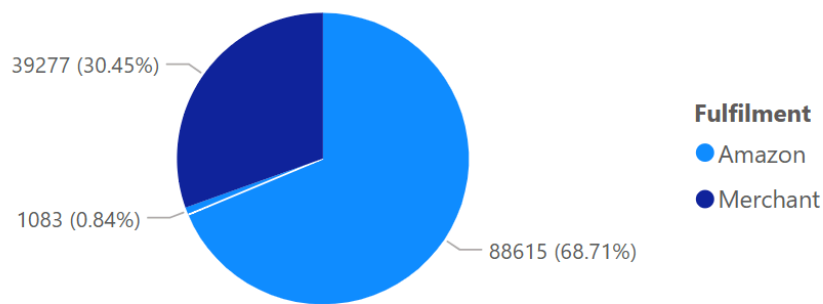
Analysis to the chart

This chart shows two lines with red and blue for each line. For for prominent graph, Jitter is adjusted at highest. Then it is clear that fulfilment almost dictate the ship service level, where Amazon fulfilment provide Expedited service which is used in most Amazon orders with a few exception. In contract, order with Merchant fulfilment exclusively used standard ship level with no exception according to chart where no red point seen among the blue area.

Confirmation from Power BI

This pie chart proves our research at Weka - Amazon monopolize the Expedited Fulfilment, while some Amazon order still use Standard

Count (Fulfilment and ship-service-level)



Fulfilment	Count
Amazon	
Expedited	88615
Standard	1083
Merchant	
Standard	39277

Conclusion of eight analysis

ASIN & SKU

- Mostly one-to-one direct correlation
- One SKU may mapped to several ASIN

Ship city & Date

- Huge distinction of number of orders among cities
- Number of orders Increase in certain cities during certain period → Regional promotion expected

Promotion ID & Date

- Most Promotion ID applied during a period and expired after then
- Some Promotion ID could be applied almost everyday

Category & Amount

- There are ranges of price related to category of good

Status & Courier status & Fulfilment

- Elements of status depends on the fulfilment of order
- One courier status relates to several potential statuses
- Most orders were shipped

Status & Quantity

- Most orders go with one quantity of good
- All orders with zero quantity is cancelled

- Most orders - no matter its quantity - were shipped

ASIN and Style

- One style corresponds to multiple ASINs
- For ASINs with the same style, there are ASINs that are obviously more popular than others

Fulfilment & Ship service level

- Expedited ship is exclusively available in Amazon fulfilment
- Those with Merchant fulfilment are utilized Standard ship
- Most orders with Amazon fulfilment utilized Expedited ship

Classification techniques applying to the dataset

Setting the predicting attribute to **Status**, the results is gotten for confusion matrix and detailed accuracy from analysis with 4 different classifiers - J48, simple cart, native bayes, KNN which varied by setting different training set size (60% and 80%)

Result of output using 8 kinds of classification

View the raw result from Weka which was posted on GitHub

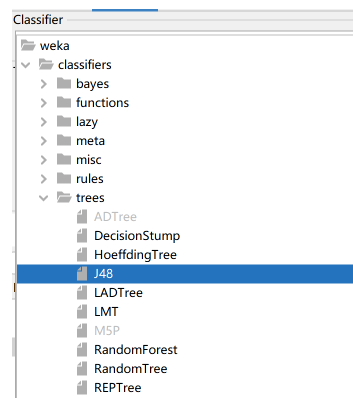
- <https://github.com/licwan62/DataAnalysis-AmazonSales/tree/main>

Summarized into data table

Classifiers	Training Set size	TP Rate (%)	FP Rate (%)	Accuracy (%)
J48 Tree	60%	97.9	0.6	97.9081
J48 Tree	80%	97.9	0.6	97.9246
Simple Cart (REP Tree)	60%	97.5	0.6	97.5017
Simple Cart (REP Tree)	80%	97.6	0.6	97.5821
Naive Bayes	60%	96.8	0.8	96.8435
Naive Bayes	80%	96.9	0.7	96.8724
K-Nearest Neighbors (IBK)	60%	89.5	7.2	89.5156
K-Nearest Neighbors (IBK)	80%	90.3	6.6	90.3326

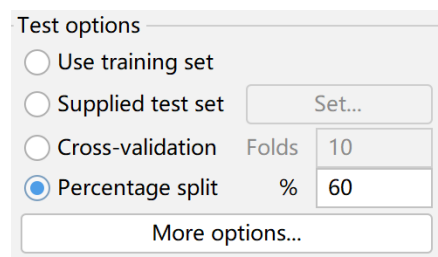
Properties explanation

- **Classifier**



- Algorithm using provided attributes of data to predict the class
- Different classifier makes different prediction while it will be verified with actual data of the class

- **Training set size**



- Percentage decided as the size of training set (60%), at the same time, left share as the size of test set (40%)
- Training set and Test set is divided part of opened data set
- Choosing division value will effect accuracy of data modelling therefore prediction

=== Confusion Matrix ===

	a	b	c	d	e	f	g	h	i	j	k	l	m	
a	4228	0	0	0	0	0	0	0	0	27	0	0	0	<-- classified as a = Cancelled
b	0	11370	0	0	0	0	0	0	0	49	0	0	0	b = Shipped - Delivered to Buyer
c	0	0	31234	0	0	0	0	0	0	0	0	0	0	c = Shipped
d	0	754	0	0	0	0	0	0	0	0	0	0	0	d = Shipped - Returned to Seller
e	0	4	0	0	0	0	0	0	0	0	0	0	0	e = Shipped - Rejected by Buyer
f	0	2	0	0	0	0	0	0	0	0	0	0	0	f = Shipped - Lost in Transit
g	0	9	0	0	0	0	0	0	6	0	0	0	0	g = Shipped - Out for Delivery
h	0	52	0	0	0	0	0	0	8	0	0	0	0	h = Shipped - Returning to Seller
i	0	58	0	0	0	0	0	0	316	0	0	0	0	i = Shipped - Picked Up
j	25	0	2	0	0	0	0	0	0	201	17	0	0	j = Pending
k	0	0	0	0	0	0	0	0	0	0	109	0	0	k = Pending - Waiting for Pick Up
l	0	1	0	0	0	0	0	0	0	0	0	0	0	l = Shipped - Damaged
m	0	0	0	0	0	0	0	0	0	0	0	0	0	m = Shipping

- **TP Rate (instance for 'a' class)**

- **TP: True Positive**

- Number of correct predictions;
 - Number within blue circle (4228) as it located at cross of actual class (vertical axis) and classified class(horizontal axis) - arranged in diagonal line

- **FN: False Negative**

- Addition of all wrong predictions for this actual class.
 - Add numbers within red rectangle (27) where it classified other than actual class

- **TP Rate** is calculated by TP and FN as $TPR = \frac{TP}{TP+FN} = \frac{4228}{4228+27} = 0.9937$

- **FP Rate (instance for 'a' class)**

- **FP: False Positive**

- Addition of all false predictions made in other actual class (if being classified as A but actual class is B,C...);
 - Add numbers without circled (25)

- **TN: True Negative**

- Addition of all true predictions made in other actual class (if being classified as B,C... and its actual class is not A);
 - True is not means correctly classified but correctly justified the whether or not this instance belongs to target class
 - Add numbers in orange rectangle
(11370+754+4+2+9+52+58+1+31234+2+49+6+8+316+201+17+109=44192)

- **FP Rate** is calculated by FP and TN as $FPR = \frac{FP}{FP+TN} = \frac{25}{25+44192} = 0.00056$

=== Detailed Accuracy By Class ===

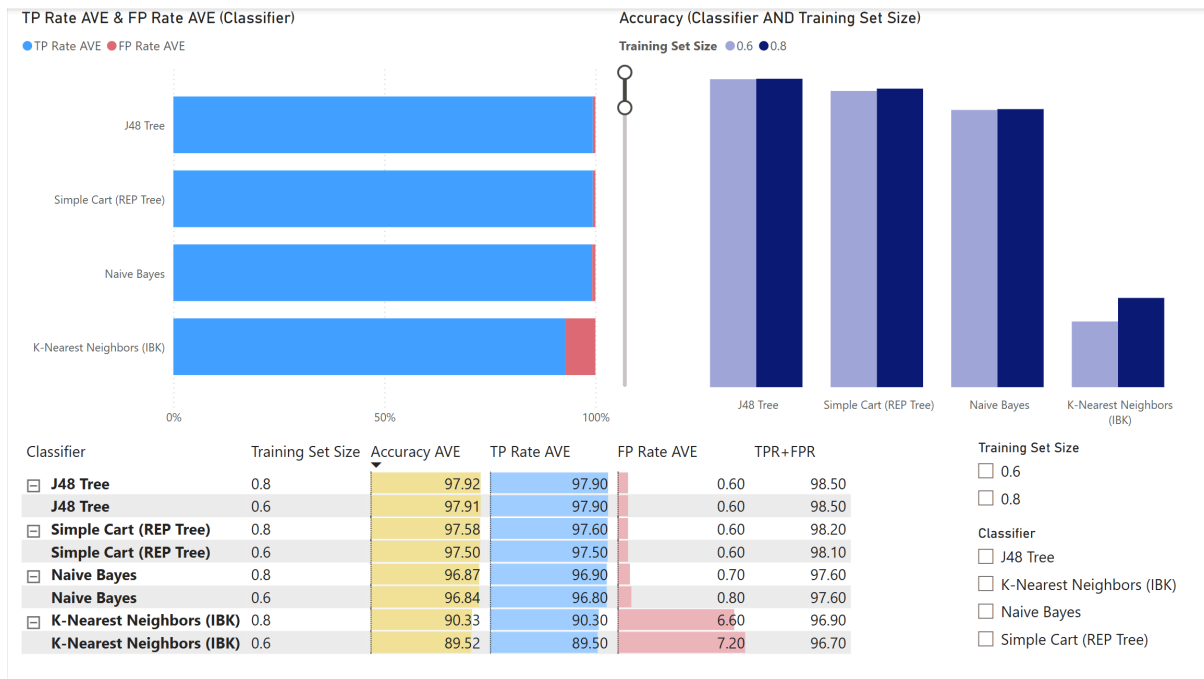
TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.994	0.001	0.994	0.994	0.994	0.993	1.000	0.998	Cancelled
0.996	0.024	0.928	0.996	0.961	0.949	0.989	0.936	Shipped - Delivered to Buyer
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Shipped
0.000	0.000	?	0.000	?	?	0.890	0.069	Shipped - Returned to Seller
0.000	0.000	?	0.000	?	?	0.491	0.000	Shipped - Rejected by Buyer
0.000	0.000	?	0.000	?	?	0.994	0.004	Shipped - Lost in Transit
0.000	0.000	?	0.000	?	?	0.925	0.018	Shipped - Out for Delivery
0.000	0.000	?	0.000	?	?	0.953	0.052	Shipped - Returning to Seller
0.845	0.001	0.834	0.845	0.839	0.838	0.985	0.896	Shipped - Picked Up
0.820	0.001	0.882	0.820	0.850	0.850	0.973	0.894	Pending
1.000	0.000	0.865	1.000	0.928	0.930	1.000	0.886	Pending - Waiting for Pick Up
0.000	0.000	?	0.000	?	?	0.500	0.000	Shipped - Damaged
?	0.000	?	?	?	?	?	?	Shipping
Weighted Avg.	0.979	0.006	?	0.979	?	?	0.995	0.967

- **Accuracy**

- Correctly classified instance in all ones

Analysis to classification statistics

Statistics visualized into graphs using Power BI



As the data table and graphs show evidently that J48 is the best classification since it result in best accuracy while KNN (IBK in Weka) get the worst accuracy.

TP Rate, Accuracy and FP Rate seems to have negative correlation but they are not added up to 100% as it seems to be.

Training set size also affect the accuracy of outcome. As the chart at top-right, same classification using 80% training set size get slightly better accuracy compared to 60%.

While larger proportion for training set is not inevitably lead to higher accuracy. There is a balance between overfitting and underfitting.

- Overfitting means the model is too complex which lead to huge loss in test set since too much insignificant detail and noise included into calculation
- Underfitting means too simple leading to both training set loss and test set loss

Therefore, it is importance to choose a clever proportion in splitting